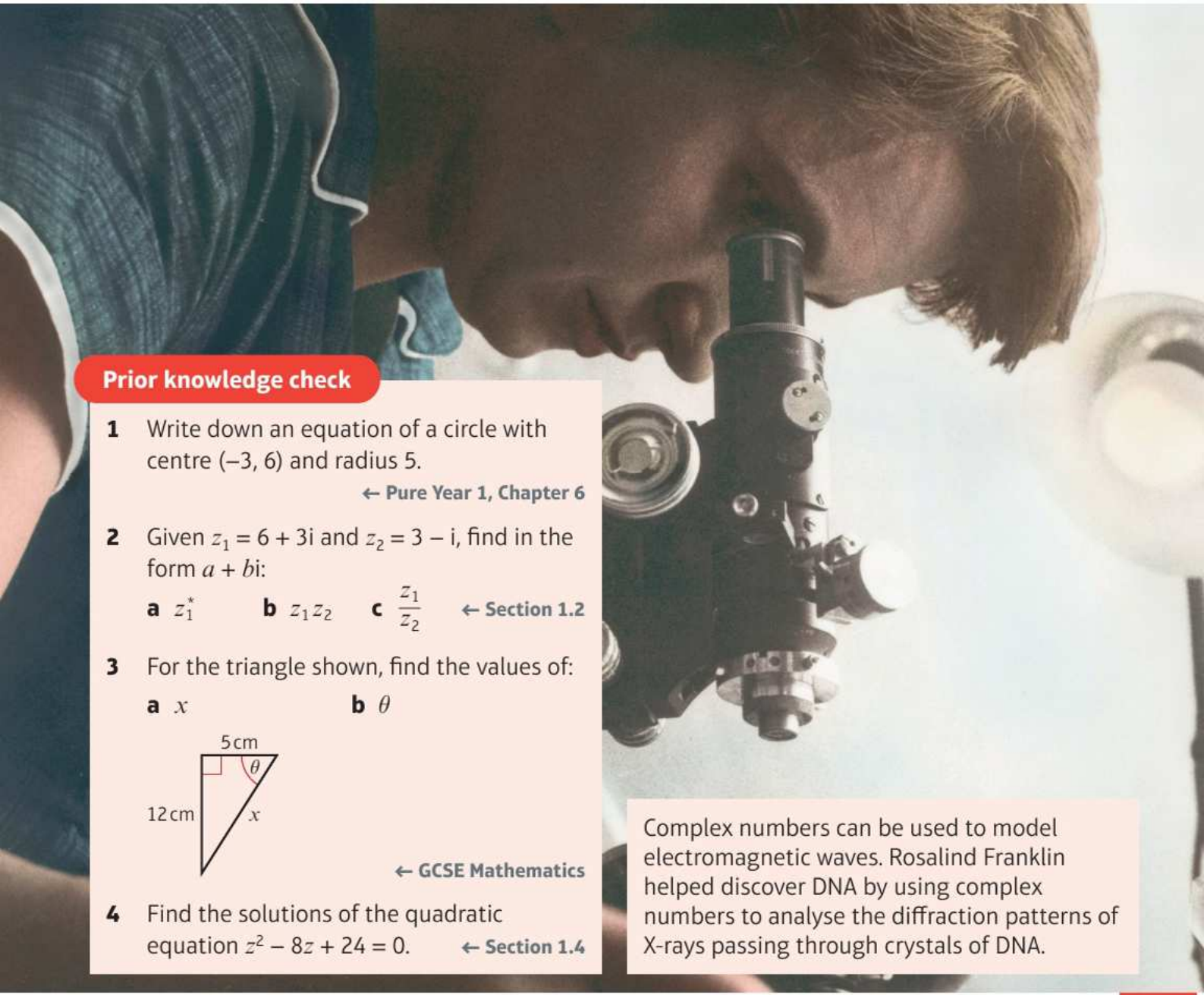
Argand diagrams

2

Objectives

After completing this chapter you should be able to:

- Show complex numbers on an Argand diagram → pages 18–19
- Find the modulus and argument of a complex number → pages 20–23
- Write a complex number in modulus-argument form → pages 23–28
- Represent loci on an Argand diagram → pages 28–36
- Represent regions on an Argand diagram → pages 36–38



Prior knowledge check

1 Write down an equation of a circle with centre (–3, 6) and radius 5.

← Pure Year 1, Chapter 6

2 Given $z_1 = 6 + 3i$ and $z_2 = 3 - i$, find in the form $a + bi$:

a z_1^* **b** $z_1 z_2$ **c** $\frac{z_1}{z_2}$ ← Section 1.2

3 For the triangle shown, find the values of:

a x **b** θ

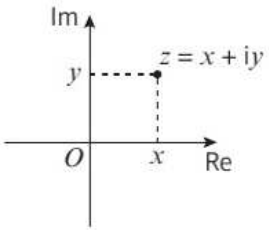
← GCSE Mathematics

4 Find the solutions of the quadratic equation $z^2 - 8z + 24 = 0$. ← Section 1.4

Complex numbers can be used to model electromagnetic waves. Rosalind Franklin helped discover DNA by using complex numbers to analyse the diffraction patterns of X-rays passing through crystals of DNA.

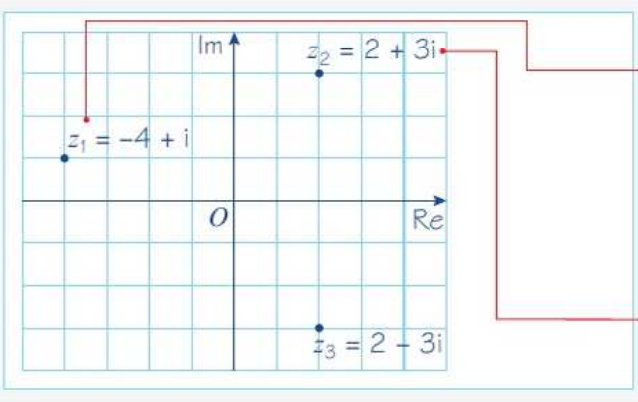
2.1 Argand diagrams

■ **You can represent complex numbers on an Argand diagram. The x-axis on an Argand diagram is called the real axis and the y-axis is called the imaginary axis. The complex number $z = x + iy$ is represented on the diagram by the point $P(x, y)$, where x and y are Cartesian coordinates.**



Example 1

Show the complex numbers $z_1 = -4 + i$, $z_2 = 2 + 3i$ and $z_3 = 2 - 3i$ on an Argand diagram.



The real part of each number describes its horizontal position, and the imaginary part describes its vertical position. For example, $z_1 = -4 + i$ has real part -4 and imaginary part 1.

Note that z_2 and z_3 are complex conjugates. On an Argand diagram, complex conjugate pairs are symmetrical about the real axis. ← Section 1.3

Complex numbers can also be represented as vectors on the Argand diagram.

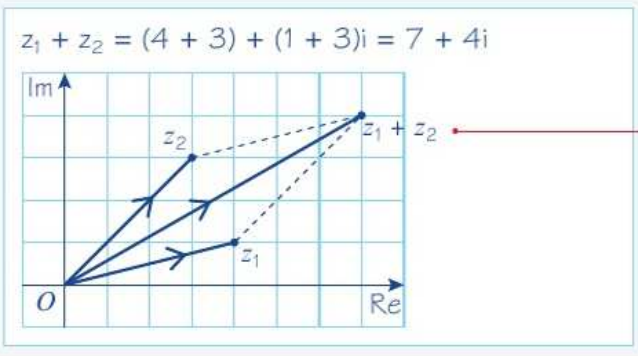
■ **The complex number $z = x + iy$ can be represented as the vector $\begin{pmatrix} x \\ y \end{pmatrix}$ on an Argand diagram.**

You can add or subtract complex numbers on an Argand diagram by adding or subtracting their corresponding vectors.

Example 2

$z_1 = 4 + i$ and $z_2 = 3 + 3i$. Show z_1, z_2 and $z_1 + z_2$ on an Argand diagram.

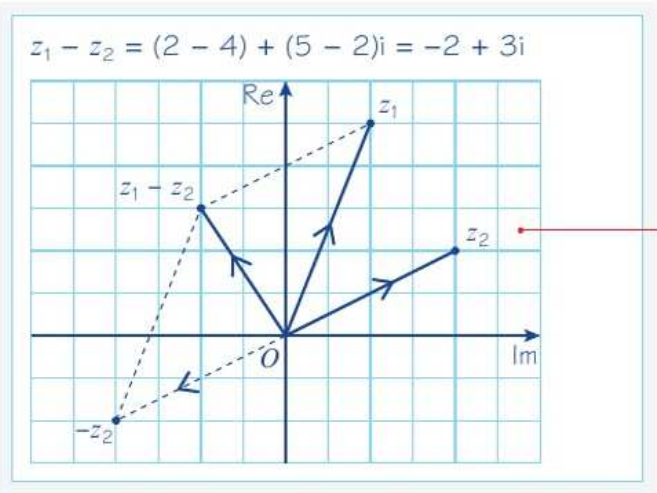
$z_1 + z_2 = (4 + 3) + (1 + 3)i = 7 + 4i$



The vector representing $z_1 + z_2$ is the diagonal of the parallelogram with vertices at O, z_1 and z_2. You can use vector addition to find $z_1 + z_2$:
$\begin{pmatrix} 4 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$

Example 3

$z_1 = 2 + 5i$ and $z_2 = 4 + 2i$. Show z_1, z_2 and $z_1 - z_2$ on an Argand diagram.



The vector corresponding to z_2 is $\begin{pmatrix}4\\2\end{pmatrix}$, so the vector corresponding to $-z_2$ is $\begin{pmatrix}-4\\-2\end{pmatrix}$.

The vector representing $z_1 - z_2$ is the diagonal of the parallelogram with vertices at O, z_1 and $-z_2$.

Online Explore adding and subtracting complex numbers on an Argand diagram using GeoGebra.

Exercise 2A

1 Show these numbers on an Argand diagram.

a $7 + 2\text{i}$ **b** $5 - 4\text{i}$ **c** $-6 - \text{i}$ **d** $-2 + 5\text{i}$

e 3i **f** $\sqrt{2} + 2\text{i}$ **g** $-\frac{1}{2} + \frac{5}{2}\text{i}$ **h** -4

2 $z_1 = 11 + 2\text{i}$ and $z_2 = 2 + 4\text{i}$. Show z_1, z_2 and $z_1 + z_2$ on an Argand diagram.

3 $z_1 = -3 + 6\text{i}$ and $z_2 = 8 - \text{i}$. Show z_1, z_2 and $z_1 + z_2$ on an Argand diagram.

4 $z_1 = 8 + 4\text{i}$ and $z_2 = 6 + 7\text{i}$. Show z_1, z_2 and $z_1 - z_2$ on an Argand diagram.

5 $z_1 = -6 - 5\text{i}$ and $z_2 = -4 + 4\text{i}$. Show z_1, z_2 and $z_1 - z_2$ on an Argand diagram.

(P) **6** $z_1 = 7 - 5\text{i}$, $z_2 = a + b\text{i}$ and $z_3 = -3 + 2\text{i}$ where $a, b \in \mathbb{Z}$. Given that $z_3 = z_1 + z_2$,

a find the values of a and b **b** show z_1, z_2 and z_3 on an Argand diagram.

(P) **7** $z_1 = p + q\text{i}$, $z_2 = 9 - 5\text{i}$ and $z_3 = -8 + 5\text{i}$ where $p, q \in \mathbb{Z}$. Given that $z_3 = z_1 + z_2$,

a find the values of p and q **b** show z_1, z_2 and z_3 on an Argand diagram.

(E) **8** The solutions to the quadratic equation $z^2 - 6z + 10 = 0$ are z_1 and z_2.

a Find z_1 and z_2, giving your answers in the form $p \pm q\text{i}$, where p and q are integers. **(3 marks)**

b Show, on an Argand diagram, the points representing the complex numbers z_1 and z_2. **(2 marks)**

(E/P) **9** $\text{f}(z) = 2z^3 - 19z^2 + 64z - 60$

a Show that $\text{f}\left(\frac{3}{2}\right) = 0$. **(1 mark)**

b Use algebra to solve $\text{f}(z) = 0$ completely. **(4 marks)**

c Show all three solutions on an Argand diagram. **(2 marks)**

Challenge

a Find all the solutions to the equation $z^6 = 1$.
b Show each solution on an Argand diagram.
c Show that each solution lies on a circle with centre (0, 0) and radius 1.

Hint There will be 6 distinct roots in total. Write $z^6 = 1$ as $(z^3 - 1)(z^3 + 1) = 0$, then find three distinct roots of $z^3 - 1 = 0$ and three distinct roots of $z^3 + 1 = 0$.

2.2 Modulus and argument

The **modulus** or absolute value of a complex number is the magnitude of its corresponding vector.

- **The modulus of a complex number, $|z|$, is the distance from the origin to that number on an Argand diagram. For a complex number $z = x + iy$, the modulus is given by $|z| = \sqrt{x^2 + y^2}$.**

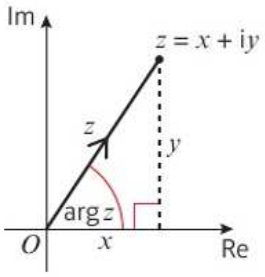
Notation The modulus of the complex number z is written as r, $|z|$ or $|x + iy|$.

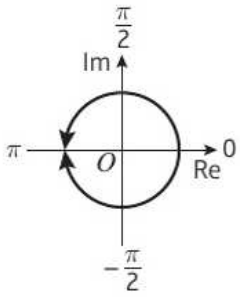
The **argument** of a complex number is the angle its corresponding vector makes with the positive real axis.

- **The argument of a complex number, arg z, is the angle between the positive real axis and the line joining that number to the origin on an Argand diagram, measured in an anticlockwise direction. For a complex number $z = x + iy$, the argument, θ, satisfies $\tan\theta = \frac{y}{x}$**

Notation The argument of the complex number z is written as arg z. It is usually given in radians, where
- 2π radians = 360°
- π radians = 180°

← Pure Year 2, Section 5.1





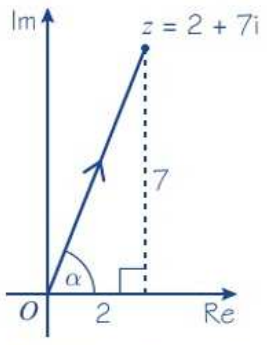
The argument θ of any complex number is usually given in the range $-\pi < \theta \leqslant \pi$. This is sometimes referred to as the **principal argument**.

Example 4

$z = 2 + 7i$, find:

a the modulus of z **b** the argument of z, giving your answer in radians to 2 decimal places.

a



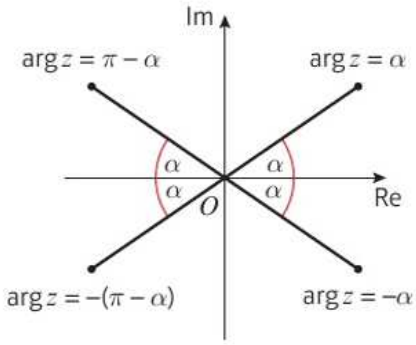
Modulus: $|z| = |2 + 7i| = \sqrt{2^2 + 7^2}$
$= \sqrt{53}$

Sketch the Argand diagram, showing the position of the number.

b Argument: $\tan\alpha = \frac{7}{2}$ $\alpha = 1.2924\ldots$ radians

arg z = 1.29 radians (2 d.p.)

If z does not lie in the first quadrant, you can use the Argand diagram to help you find its argument.



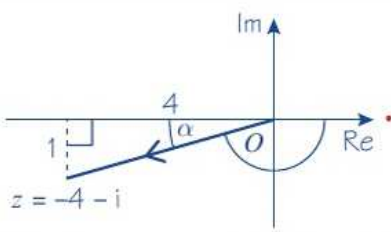
- **Let α be the positive acute angle made with the real axis by the line joining the origin and z.**
  - **If z lies in the first quadrant then arg $z = \alpha$.**
  - **If z lies in the second quadrant then arg $z = \pi - \alpha$.**
  - **If z lies in the third quadrant then arg $z = -(\pi - \alpha)$.**
  - **If z lies in the fourth quadrant then arg $z = -\alpha$.**

Example 5

$z = -4 - i$, find:

a the modulus of z **b** the argument of z, giving your answer in radians to 2 decimal places.

a



Modulus: $|z| = |-4 - i| = \sqrt{(-4)^2 + (-1)^2}$
$= \sqrt{17}$

Sketch the Argand diagram, showing the position of the number.

b Argument: $\tan\alpha = \frac{1}{4}$ $\alpha = 0.2449\ldots$ radians
arg $z = -(\pi - 0.2449)$
$= -2.90$ radians (2 d.p.)

Here z is in the third quadrant, so the required argument is $-(\pi - \alpha)$.

Exercise 2B

1 For each of the following complex numbers,

i find the modulus, writing your answer in surd form if necessary

ii find the argument, writing your answer in radians to 2 decimal places.

a $z = 12 + 5i$ **b** $z = \sqrt{3} + i$ **c** $z = -3 + 6i$

d $z = 2 - 2i$ **e** $z = -8 - 7i$ **f** $z = -4 + 11i$

g $z = 2\sqrt{3} - i\sqrt{3}$ **h** $z = -8 - 15i$

Hint In part **c**, the complex number is in the second quadrant, so the argument will be $\pi - \alpha$. In part **d**, the complex number is in the fourth quadrant, so the argument will be $-\alpha$.

2 For each of the following complex numbers,

i find the modulus, writing your answer in surd form

ii find the argument, writing your answer in terms of π.

a $2 + 2i$ **b** $5 + 5i$ **c** $-6 + 6i$ **d** $-a - ai$, $a \in \mathbb{R}$

E **3** $z = -40 - 9\text{i}$

a Show z on an Argand diagram. **(1 mark)**

b Calculate $\arg z$, giving your answer in radians to 2 decimal places. **(2 marks)**

E **4** $z = 3 + 4\text{i}$

a Show that $z^2 = -7 + 24\text{i}$. **(2 marks)**

Find, showing your working:

b $|z^2|$ **(2 marks)**

c $\arg(z^2)$, giving your answer in radians to 2 decimal places. **(2 marks)**

d Show z and z^2 on an Argand diagram. **(1 mark)**

E **5** The complex numbers z_1 and z_2 are given by $z_1 = 4 + 6\text{i}$ and $z_2 = 1 + \text{i}$.

Find, showing your working:

a $\frac{z_1}{z_2}$ in the form $a + b\text{i}$, where a and b are real **(3 marks)**

b $\left|\frac{z_1}{z_2}\right|$ **(2 marks)**

c $\arg\frac{z_1}{z_2}$, giving your answer in radians to 2 decimal places. **(2 marks)**

E/P **6** The complex numbers z_1 and z_2 are such that $z_1 = 3 + 2p\text{i}$ and $\frac{z_1}{z_2} = 1 - \text{i}$ where p is a real constant.

a Find z_2 in the form $a + b\text{i}$, giving the real numbers a and b in terms of p. **(3 marks)**

Given that $\arg z_2 = \tan^{-1} 5$,

b find the value of p **(2 marks)**

c find the value of $|z_2|$ **(2 marks)**

d show z_1, z_2 and $\frac{z_1}{z_2}$ on a single Argand diagram. **(2 marks)**

E **7** $z = \frac{26}{2 - 3\text{i}}$, find:

a z in the form $a + \text{i}b$ where $a, b \in \mathbb{R}$ **(2 marks)**

b z^2 in the form $a + \text{i}b$ where $a, b \in \mathbb{R}$ **(2 marks)**

c $|z|$ **(2 marks)**

d $\arg(z^2)$, giving your answer in radians to 2 decimal places. **(2 marks)**

E/P **8** $z_1 = 4 + 2\text{i}$, $z_2 = 2 + 4\text{i}$, $z_3 = a + b\text{i}$ where $a, b \in \mathbb{R}$.

a Find the exact value of $|z_1 + z_2|$. **(2 marks)**

Given that $w = \frac{z_1 z_3}{z_2}$,

b find w in terms of a and b, giving your answer in the form $x + \text{i}y$, $x, y \in \mathbb{R}$. **(4 marks)**

Given also that $w = \frac{21}{5} - \frac{22}{5}\text{i}$, find:

c the values of a and b **(3 marks)**

d $\arg w$, giving your answer in radians to 2 decimal places. **(2 marks)**

E/P **9** The complex number w is given by $w = 6 + 3i$. Find:

a $|w|$ **(1 mark)**

b $\arg w$, giving your answer in radians to 2 decimal places. **(2 marks)**

Given that $\arg(\lambda + 5i + w) = \frac{\pi}{4}$ where λ is a real constant,

c find the value of λ. **(2 marks)**

E **10** $z = -1 - i\sqrt{3}$, find:

a $|z|$ **(1 mark)**

b $\left|\frac{z}{z^*}\right|$ **(4 marks)**

c $\arg z$, $\arg(z^*)$ and $\arg\frac{z}{z^*}$, giving your answers in terms of π. **(3 marks)**

E/P **11** The complex numbers w and z are given by $w = k + i$ and $z = -4 + 5ki$, where k is a real constant. Given that $\arg(w + z) = \frac{2\pi}{3}$, find the exact value of k. **(6 marks)**

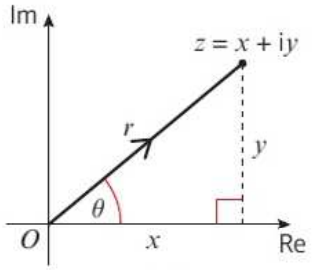
E/P **12** The complex numbers w and z are defined such that $\arg w = \frac{\pi}{10}$, $|w| = 5$ and $\arg z = \frac{2\pi}{5}$

Given that $\arg(w + z) = \frac{\pi}{5}$, find the value of $|z|$. **(4 marks)**

2.3 Modulus–argument form of complex numbers

You can write any complex number in terms of its modulus and argument.

- **For a complex number z with $|z| = r$ and $\arg z = \theta$, the modulus–argument form of z is $z = r(\cos\theta + i\sin\theta)$**



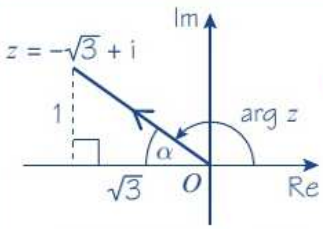
From the right-angled triangle, $x = r\cos\theta$ and $y = r\sin\theta$.

$z = x + iy = r\cos\theta + ir\sin\theta = r(\cos\theta + i\sin\theta)$

This formula works for a complex number in any quadrant of the Argand diagram. The argument, θ, is usually given in the range $-\pi < \theta \leqslant \pi$, although the formula works for any value of θ measured anticlockwise from the positive real axis.

Example 6

Express $z = -\sqrt{3} + i$ in the form $r(\cos\theta + i\sin\theta)$, where $-\pi < \theta \leqslant \pi$.



Sketch the Argand diagram, showing the position of the number.

Here z is in the second quadrant, so the required argument is $\pi - \alpha$.

$r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$

$\theta = \arg z = \pi - \arctan\left(\frac{1}{\sqrt{3}}\right) = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$

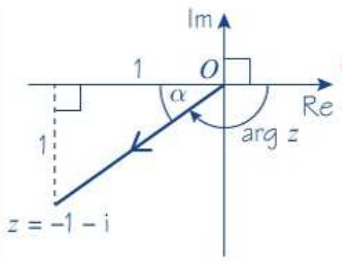
Find r and θ.

Therefore, $z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

Apply $z = r(\cos\theta + i\sin\theta)$.

Example 7

Express $z = -1 - \mathrm{i}$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $-\pi < \theta \leqslant \pi$.



Sketch the Argand diagram, showing the position of the number.

Here z is in the third quadrant, so the required argument is $-(\pi - \alpha)$.

$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$

$\theta = \arg z = -\pi + \arctan\left(\frac{1}{1}\right) = -\pi + \frac{\pi}{4} = -\frac{3\pi}{4}$

Find r and θ.

Therefore, $z = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{3\pi}{4}\right)\right)$

Apply $z = r(\cos\theta + \mathrm{i}\sin\theta)$.

Exercise 2C

1 Express the following in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $-\pi < \theta \leqslant \pi$. Give the exact values of r and θ where possible, or values to 2 decimal places otherwise.

a $2 + 2\mathrm{i}$ **b** $3\mathrm{i}$ **c** $-3 + 4\mathrm{i}$ **d** $1 - \sqrt{3\mathrm{i}}$

e $-2 - 5\mathrm{i}$ **f** -20 **g** $7 - 24\mathrm{i}$ **h** $-5 + 5\mathrm{i}$

2 Express these in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, giving exact values of r and θ where possible, or values to two decimal places otherwise.

a $\frac{3}{1 + \mathrm{i}\sqrt{3}}$ **b** $\frac{1}{2 - \mathrm{i}}$ **c** $\frac{1 + \mathrm{i}}{1 - \mathrm{i}}$

3 Express the following in the form $x + \mathrm{i}y$, where $x, y \in \mathbb{R}$.

a $5\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$ **b** $\frac{1}{2}\left(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6}\right)$ **c** $6\left(\cos\frac{5\pi}{6} + \mathrm{i}\sin\frac{5\pi}{6}\right)$

d $3\left(\cos\left(-\frac{2\pi}{3}\right) + \mathrm{i}\sin\left(-\frac{2\pi}{3}\right)\right)$ **e** $2\sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$ **f** $-4\left(\cos\frac{7\pi}{6} + \mathrm{i}\sin\frac{7\pi}{6}\right)$

(E) **4 a** Express the complex number $z = 4\left(\cos\left(\frac{2\pi}{3}\right) + i\sin\left(\frac{2\pi}{3}\right)\right)$ in the form $x + \mathrm{i}y$, where $x, y \in \mathbb{R}$. **(2 marks)**

b Show the complex number z on an Argand diagram. **(1 mark)**

(E) **5** The complex number z is such that $|z| = 7$ and $\arg z = \frac{11\pi}{6}$. Find z in the form $p + qi$, where p and q are exact real numbers to be found. **(3 marks)**

(E) **6** The complex number z is such that $|z| = 5$ and $\arg z = -\frac{4\pi}{3}$. Find z in the form $a + bi$, where a and b are exact real numbers to be found. **(3 marks)**

You can use the following rules to multiply complex numbers quickly when they are given in modulus–argument form.

■ **For any two complex numbers z_1 and z_2,**

- $|z_1 z_2| = |z_1||z_2|$
- $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

Note You multiply the moduli and add the arguments.

To prove these results, consider z_1 and z_2 in modulus–argument form:

$$z_1 = r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1) \text{ and } z_2 = r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)$$

Multiplying these numbers together, you get

$$\begin{aligned} z_1 z_2 &= r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1) \times r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2) \\ &= r_1 r_2(\cos\theta_1 + \mathrm{i}\sin\theta_1)(\cos\theta_2 + \mathrm{i}\sin\theta_2) \\ &= r_1 r_2(\cos\theta_1\cos\theta_2 + \mathrm{i}\cos\theta_1\sin\theta_2 + \mathrm{i}\sin\theta_1\cos\theta_2 + \mathrm{i}^2\sin\theta_1\sin\theta_2) \\ &= r_1 r_2(\cos\theta_1\cos\theta_2 + \mathrm{i}\cos\theta_1\sin\theta_2 + \mathrm{i}\sin\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) \\ &= r_1 r_2((\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + \mathrm{i}(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)) \\ &= r_1 r_2(\cos(\theta_1 + \theta_2) + \mathrm{i}\sin(\theta_1 + \theta_2)) \end{aligned}$$

Links The last step of this working makes use of the trigonometric addition formulae:
$\sin(A \pm B) \equiv \sin A\cos B \pm \cos A\sin B$
$\cos(A \pm B) \equiv \cos A\cos B \mp \sin A\sin B$
← Pure Year 2, Section 7.1

This complex number is in modulus–argument form, with modulus $r_1 r_2$ and argument $\theta_1 + \theta_2$, as required.

You can derive similar results for dividing two complex numbers given in modulus–argument form.

■ **For any two complex numbers z_1 and z_2,**

- $\left|\dfrac{z_1}{z_2}\right| = \dfrac{|z_1|}{|z_2|}$
- $\arg\left(\dfrac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$

Note You divide the moduli and subtract the arguments.

To prove these results, again consider z_1 and z_2 in modulus–argument form:

$z_1 = r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)$

Online Explore multiplying and dividing complex numbers on an Argand diagram using GeoGebra.



Dividing z_1 by z_2 you get

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1)}{r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)} \\ &= \frac{r_1(\cos\theta_1 + \mathrm{i}\sin\theta_1)}{r_2(\cos\theta_2 + \mathrm{i}\sin\theta_2)} \times \frac{(\cos\theta_2 - \mathrm{i}\sin\theta_2)}{(\cos\theta_2 - \mathrm{i}\sin\theta_2)} \\ &= \frac{r_1(\cos\theta_1\cos\theta_2 - \mathrm{i}\cos\theta_1\sin\theta_2 + \mathrm{i}\sin\theta_1\cos\theta_2 - \mathrm{i}^2\sin\theta_1\sin\theta_2)}{r_2(\cos\theta_2\cos\theta_2 - \mathrm{i}\cos\theta_2\sin\theta_2 + \mathrm{i}\sin\theta_2\cos\theta_2 - \mathrm{i}^2\sin\theta_2\sin\theta_2)} \\ &= \frac{r_1((\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) + \mathrm{i}(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2))}{r_2(\cos^2\theta_2 + \sin^2\theta_2)} \\ &= \frac{r_1}{r_2}(\cos(\theta_1 - \theta_2) + \mathrm{i}\sin(\theta_1 - \theta_2)) \end{aligned}$$

Links The last step of this working makes use of the trigonometric addition formulae together with the identity $\sin^2\theta + \cos^2\theta \equiv 1$
← Pure Year 1, Section 10.3

This complex number is in modulus–argument form, with modulus $\dfrac{r_1}{r_2}$ and argument $\theta_1 - \theta_2$, as required.

Example 8

$z_1 = 3\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$ and $4\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

a Find: **i** $|z_1 z_2|$ **ii** $\arg(z_1 z_2)$

b Hence write $z_1 z_2$ in the form: **i** $r(\cos\theta + i\sin\theta)$ **ii** $x + iy$

a i $|z_1 z_2| = |z_1||z_2|$

$|z_1| = 3,\ |z_2| = 4$

$|z_1 z_2| = 3 \times 4 = 12$

For a complex number in the form $z = r(\cos\theta + i\sin\theta)$, $|z| = r$

ii $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

$\arg z_1 = \frac{5\pi}{12},\ \arg z_2 = \frac{\pi}{12}$

For a complex number in the form $z = r(\cos\theta + i\sin\theta)$, $\arg z = \theta$

$\arg(z_1 z_2) = \frac{5\pi}{12} + \frac{\pi}{12} = \frac{6\pi}{12} = \frac{\pi}{2}$

b i $z_1 z_2 = r(\cos\theta + i\sin\theta) = 12\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

$z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$

ii $\cos\frac{\pi}{2} = 0,\ \sin\frac{\pi}{2} = 1$

Evaluate $\cos\frac{\pi}{2}$ and $\sin\frac{\pi}{2}$

$z_1 z_2 = 12(0 + i(1)) = 12i$

Example 9

$z_1 = 2\left(\cos\frac{\pi}{15} + i\sin\frac{\pi}{15}\right)$ and $z_2 = 3\left(\cos\frac{2\pi}{5} - i\sin\frac{2\pi}{5}\right)$

Express $z_1 z_2$ in the form $x + iy$.

Rewrite z_2 in the form $z_2 = r(\cos\theta + i\sin\theta)$:

Watch out z_2 is not initially given in modulus–argument form.

$\cos\left(-\frac{2\pi}{5}\right) = \cos\frac{2\pi}{5}$ and $\sin\left(-\frac{2\pi}{5}\right) = -\sin\frac{2\pi}{5}$

Use $\cos(-\theta) = \cos\theta$ and $\sin(-\theta) = -\sin\theta$

$z_2 = 3\left(\cos\left(-\frac{2\pi}{5}\right) + i\sin\left(-\frac{2\pi}{5}\right)\right)$

z_2 is now in modulus–argument form.

$z_1 z_2 = 2\left(\cos\frac{\pi}{15} + i\sin\frac{\pi}{15}\right) \times 3\left(\cos\left(-\frac{2\pi}{5}\right) + i\sin\left(-\frac{2\pi}{5}\right)\right)$

$= 2 \times 3\left(\cos\left(\frac{\pi}{15} - \frac{2\pi}{5}\right) + i\sin\left(\frac{\pi}{15} - \frac{2\pi}{5}\right)\right)$

Apply the result $z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$

$= 6\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$

$= 6\left(\frac{1}{2} + i\left(-\frac{\sqrt{3}}{2}\right)\right)$

Apply $\cos\left(-\frac{\pi}{3}\right) = \frac{1}{2}$ and $\sin\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$

$= 3 - 3\sqrt{3}i$

Example 10

Express $\dfrac{\sqrt{2}\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)}{2\left(\cos\frac{5\pi}{6}+\mathrm{i}\sin\frac{5\pi}{6}\right)}$ in the form $x+\mathrm{i}y$.

$$\frac{\sqrt{2}\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)}{2\left(\cos\frac{5\pi}{6}+\mathrm{i}\sin\frac{5\pi}{6}\right)}$$

$$=\frac{\sqrt{2}}{2}\left(\cos\left(\frac{\pi}{12}-\frac{5\pi}{6}\right)+\mathrm{i}\sin\left(\frac{\pi}{12}-\frac{5\pi}{6}\right)\right)$$

$$=\frac{\sqrt{2}}{2}\left(\cos\left(-\frac{3\pi}{4}\right)+\mathrm{i}\sin\left(-\frac{3\pi}{4}\right)\right)$$

$$=\frac{\sqrt{2}}{2}\left(-\frac{1}{\sqrt{2}}+\mathrm{i}\left(-\frac{1}{\sqrt{2}}\right)\right)$$

$$=-\frac{1}{2}-\frac{1}{2}\mathrm{i}$$

Both numbers are in modulus–argument form, so you can divide the moduli and subtract the arguments.

Simplify.

Apply $\cos\left(-\frac{3\pi}{4}\right)=-\frac{1}{\sqrt{2}}$ and $\sin\left(-\frac{3\pi}{4}\right)=-\frac{1}{\sqrt{2}}$

Exercise 2D

1 For each given z_1 and z_2, find the following in the form $r(\cos\theta+\mathrm{i}\sin\theta)$:

i $|z_1z_2|$ **ii** $\arg(z_1z_2)$ **iii** z_1z_2

a $z_1=5\left(\cos\frac{3\pi}{8}+\mathrm{i}\sin\frac{3\pi}{8}\right)$, $z_2=6\left(\cos\frac{7\pi}{8}+\mathrm{i}\sin\frac{7\pi}{8}\right)$

b $z_1=\sqrt{2}\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$, $z_2=4\sqrt{2}\left(\cos\frac{3\pi}{4}+\mathrm{i}\sin\frac{3\pi}{4}\right)$

2 Given $z_1=8\left(\cos\frac{8\pi}{5}+\mathrm{i}\sin\frac{8\pi}{5}\right)$ and $z_2=4\left(\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}\right)$, write down the modulus and argument of:

a z_1z_2 **b** $\frac{z_1}{z_2}$ **c** z_1^2

3 Express the following in the form $x+\mathrm{i}y$:

a $(\cos2\theta+\mathrm{i}\sin2\theta)(\cos3\theta+\mathrm{i}\sin3\theta)$

b $\left(\cos\frac{3\pi}{11}+\mathrm{i}\sin\frac{3\pi}{11}\right)\left(\cos\frac{8\pi}{11}+\mathrm{i}\sin\frac{8\pi}{11}\right)$

c $3\left(\cos\frac{\pi}{4}+\mathrm{i}\sin\frac{\pi}{4}\right)\times2\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)$

d $\sqrt{6}\left(\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}\right)\times\sqrt{3}\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)$

e $4\left(\cos\frac{5\pi}{9}-\mathrm{i}\sin\frac{5\pi}{9}\right)\times\frac{1}{2}\left(\cos\frac{5\pi}{18}-\mathrm{i}\sin\frac{5\pi}{18}\right)$

f $6\left(\cos\frac{\pi}{10}+\mathrm{i}\sin\frac{\pi}{10}\right)\times5\left(\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}\right)\times\frac{1}{3}\left(\cos\frac{2\pi}{5}+\mathrm{i}\sin\frac{2\pi}{5}\right)$

g $(\cos4\theta+\mathrm{i}\sin4\theta)(\cos\theta-\mathrm{i}\sin\theta)$

h $3\left(\cos\frac{\pi}{12}+\mathrm{i}\sin\frac{\pi}{12}\right)\times\sqrt{2}\left(\cos\frac{\pi}{3}-\mathrm{i}\sin\frac{\pi}{3}\right)$

Hint First make sure both numbers are in modulus–argument form.

4 Express the following in the form $x + iy$:

a $\dfrac{\cos 5\theta + i\sin 5\theta}{\cos 2\theta + i\sin 2\theta}$ **b** $\dfrac{\sqrt{2}\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)}{\frac{1}{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)}$ **c** $\dfrac{3\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)}{4\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)}$ **d** $\dfrac{\cos 2\theta - i\sin 2\theta}{\cos 3\theta + i\sin 3\theta}$

(E) **5** $z = -9 + 3i\sqrt{3}$

a Express z in the form $r(\cos\theta + i\sin\theta)$, $-\pi < \theta \leqslant \pi$ **(2 marks)**

b Given that $|w| = \sqrt{3}$ and $\arg w = \dfrac{7\pi}{12}$, express in the form $r(\cos\theta + i\sin\theta)$, $-\pi < \theta \leqslant \pi$:

i w **ii** zw **iii** $\dfrac{z}{w}$ **(4 marks)**

Challenge

By writing $z = 1 + i\sqrt{3}$ in modulus-argument form, show that:

a $z^7 = kz$ **b** $z^4 = pz$

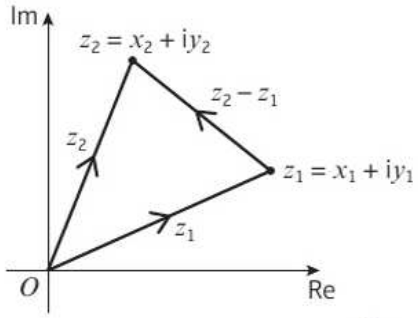
where k and p are real constants to be found.

c Show z, z^7 and z^4 on an Argand diagram and describe the geometrical relationship between them.

2.4 Loci in the Argand diagram

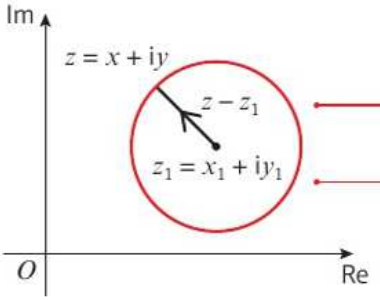
Complex numbers can be used to represent a locus of points on an Argand diagram.

■ **For two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, $|z_2 - z_1|$ represents the distance between the points z_1 and z_2 on an Argand diagram.**



Using the above result, you can replace z_2 with the general point z. The locus of points described by $|z - z_1| = r$ is a circle with centre (x_1, y_1) and radius r.

Online Explore the locus of z, when $|z - z_1| = r$, using GeoGebra.



Locus of points.

Every point z, on the circumference of the circle, is a distance of r from the centre of the circle.

☒ **Given $z_1 = x_1 + iy_1$, the locus of point z on an Argand diagram such that $|z - z_1| = r$, or $|z - (x_1 + iy_1)| = r$, is a circle with centre (x_1, y_1) and radius r.**

You can derive a Cartesian form of the equation of a circle from this form by squaring both sides:

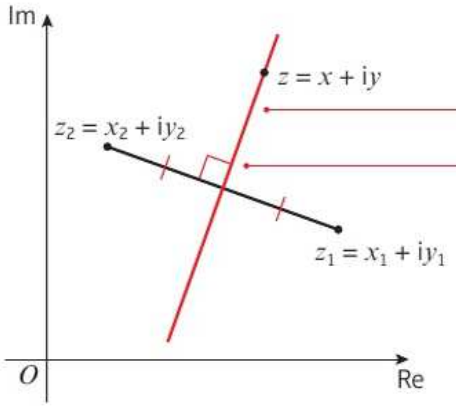
$$|z - z_1| = r$$
$$|(x - x_1) + i(y - y_1)| = r$$
$$(x - x_1)^2 + (y - y_1)^2 = r^2$$

Since $|p + qi| = \sqrt{p^2 + q^2}$

Links The Cartesian equation of a circle with centre (a, b) and radius r is $(x - a)^2 + (y - b)^2 = r^2$

← Pure Year 1, Section 6.2

The locus of points that are an equal distance from two different points z_1 and z_2 is the perpendicular bisector of the line segment joining the two points.



Locus of points.

Every point z on the line is an equal distance from points z_1 and z_2.

Online Explore the locus of z, when $|z - z_1| = |z - z_2|$, using GeoGebra.



■ **Given $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, the locus of points z on an Argand diagram such that $|z - z_1| = |z - z_2|$ is the perpendicular bisector of the line segment joining z_1 and z_2.**

Example 11

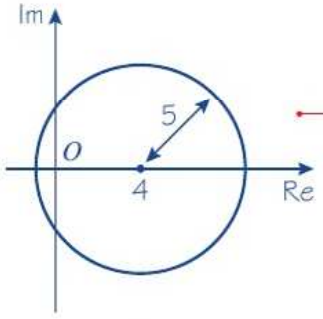
Given that z satisfies $|z - 4| = 5$,

a sketch the locus of z on an Argand diagram.

b Find the values of z that satisfy:

i both $|z - 4| = 5$ and $\text{Im}(z) = 0$ **ii** both $|z - 4| = 5$ and $\text{Re}(z) = 0$

a $|z - 4| = 5$ is a circle with centre (4, 0) and radius 5.



$|z - (x_1 + iy_1)| = r$ is represented by a circle with centre (x_1, y_1) and radius r.

Sketch a circle with centre (4, 0) and radius 5 on an Argand diagram.

b i $\text{Im}(z) = 0$ represents the real axis.
The points where the circle cuts the real axis are (−1, 0) and (9, 0).
The values of z at these points are $z = -1$ and $z = 9$.

Centre of circle is (4, 0) and radius is 5.
So consider 4 + 5 = 9 and 4 − 5 = −1.

Watch out Give your answers as complex numbers, not as coordinates.

ii $|z - 4| = 5 \Rightarrow (x - 4)^2 + y^2 = 5^2$

$(0 - 4)^2 + y^2 = 5^2$

$16 + y^2 = 25$

$y^2 = 9$

$y = \pm 3$

The points where the circle cuts the real axis are (0, 3) and (0, −3).

The values of z are $z = 3i$ and $z = -3i$.

This is the Cartesian equation of a circle with centre (4, 0) and radius 5.

$\text{Re}(z) = 0$ for all points on the imaginary axis, so set $x = 0$.

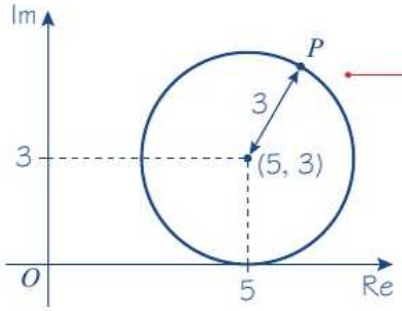
Example 12

A complex number z is represented by the point P in the Argand diagram. Given that $|z - 5 - 3i| = 3$,

a sketch the locus of P **b** find the Cartesian equation of this locus

c find the maximum value of $\arg z$ in the interval $(-\pi, \pi)$.

a

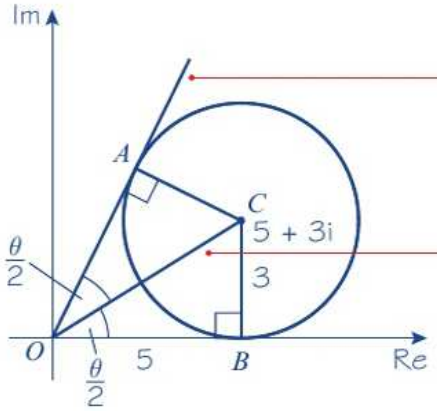


$|z - 5 - 3i|$ can be written as $|z - (5 + 3i)|$. As this distance is always equal to 3, the locus of P is a circle centre (5, 3), radius 3.

b The Cartesian equation of the locus is $(x - 5)^2 + (y - 3)^2 = 9$

The standard Cartesian equation of a circle is $(x - a)^2 + (y - b)^2 = r^2$

c



The maximum value of $\arg z$ is the angle OA makes with the positive real axis.

The line OC bisects the angle AOB.

Problem-solving

When solving geometrical problems like this one, it is helpful to draw an Argand diagram. The maximum value of $\arg(z)$ occurs when the line between the origin and P is a tangent to the circle.

Using triangle OBC:

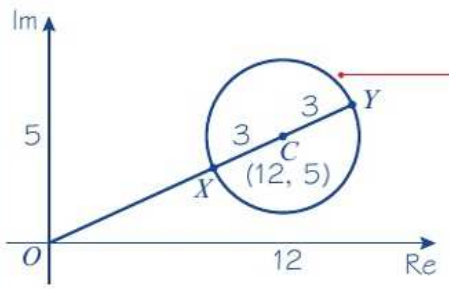
$\tan\left(\frac{\theta}{2}\right) = \frac{3}{5}$

$\theta = 2\arctan\left(\frac{3}{5}\right) = 1.08$ rad (3 s.f.)

Use circle properties. OB is perpendicular to BC, and triangles OBC and OAC are congruent.

Example 13

Given that the complex number $z = x + iy$ satisfies the equation $|z - 12 - 5i| = 3$, find the minimum value of $|z|$ and maximum value of $|z|$.



$|z|_{min} = OC - CX = 13 - 3 = 10.$

$|z|_{max} = OC + CY = 13 + 3 = 16.$

The minimum value of $|z|$ is 10 and the maximum value of $|z|$ is 16.

The locus of z is a circle centre $C(12, 5)$, radius 3.

$|z|$ represents the distance from the origin to any point on this locus.

$|z|_{min}$ and $|z|_{max}$ are represented by the distances OX and OY respectively.

The distance $OC = \sqrt{12^2 + 5^2} = 13$.

The radius $r = CX = CY = 3$.

Example 14

Given that $|z - 3| = |z + i|$,

a sketch the locus of z and find the Cartesian equation of this locus

b find the least possible value of $|z|$.

a $|z - 3| = |z + i|$ is the perpendicular bisector of the line segment joining the points (3, 0) and (0, −1).

The gradient of the line joining (0, −1) and (3, 0) is $\frac{1}{3}$

So, the gradient of the perpendicular bisector is −3.

The midpoint of the line joining (0, −1) and (3, 0) is $\left(\frac{3}{2}, -\frac{1}{2}\right)$.

$y - y_1 = m(x - x_1)$

$y + \frac{1}{2} = -3\left(x - \frac{3}{2}\right)$

$y + \frac{1}{2} = -3x + \frac{9}{2}$

$y = -3x + 4$

The locus of points z satisfying $|z - z_1| = |z - z_2|$ is the perpendicular bisector of the line segment joining z_1 to z_2.

The perpendicular bisector will pass through the midpoint.

Substitute $(x_1, y_1) = \left(\frac{3}{2}, -\frac{1}{2}\right)$ and $m = -3$ into the equation of a straight line.

Problem-solving

You could also square both sides of $|z - 3| = |z + i|$:

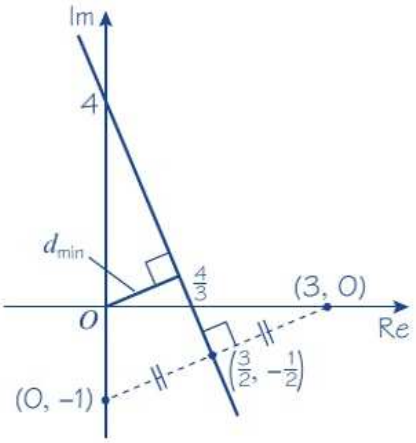
$|x + iy - 3| = |x + iy + i|$

$|(x - 3) + iy| = |x + i(y + 1)|$

$(x - 3)^2 + y^2 = x^2 + (y + 1)^2$

$x^2 - 6x + 9 + y^2 = x^2 + y^2 + 2y + 1$

$y = -3x + 4$



b The gradient of the line labelled d_{min} is $\frac{1}{3}$

The equation of this line is $y = \frac{1}{3}x$

$\frac{1}{3}x = -3x + 4$

$\frac{10}{3}x = 4$

$x = \frac{6}{5} \Rightarrow y = \frac{2}{5}$

$d_{min} = \sqrt{\left(\frac{6}{5}\right)^2 + \left(\frac{2}{5}\right)^2}$

$= \frac{2\sqrt{10}}{5}$

Problem-solving

The minimum distance is the perpendicular distance from O to the perpendicular bisector.

The line is parallel to the line joining (0, –1) and (3, 0).

The line passes through the origin.

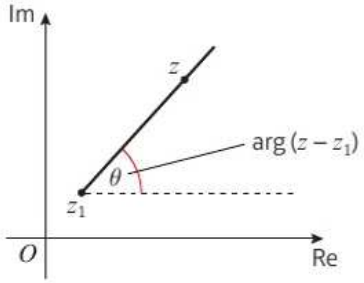
Find the point where this line intersects the perpendicular bisector.

Solve to find x and substitute into $y = \frac{1}{3}x$ to find y.

Use Pythagoras' theorem.

Locus questions can also make use of the geometric property of the argument.

- **Given $z_1 = x_1 + iy_1$, the locus of points z on an Argand diagram such that $\arg(z - z_1) = \theta$ is a half-line from, but not including, the fixed point z_1 making an angle θ with a line from the fixed point z_1 parallel to the real axis.**



Notation A **half-line** is a straight line extending from a point infinitely in one direction only.

Online Explore the locus of z, when $\arg(z - z_1) = \theta$, using GeoGebra.



You can find the Cartesian equation of the half-line corresponding to $\arg(z - z_1) = \theta$ by considering how the argument is calculated:

$$\arg(z - z_1) = \theta$$

$$\arg((x - x_1) + i(y - y_1)) = \theta$$

$$\frac{y - y_1}{x - x_1} = \tan\theta$$

$$y - y_1 = \tan\theta(x - x_1)$$

θ is a fixed angle so $\tan\theta$ is a constant.

This is the equation of a straight line with gradient $\tan\theta$ passing through the point (x_1, y_1).

Example 15

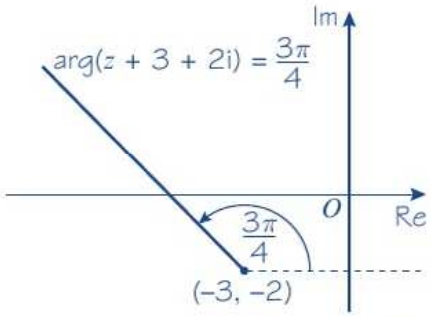
Given that $\arg(z + 3 + 2i) = \frac{3\pi}{4}$,

a sketch the locus of z on an Argand diagram

b find the Cartesian equation of the locus

c find the complex number z that satisfies both $|z + 3 + 2i| = 10$ and $\arg(z + 3 + 2i) = \frac{3\pi}{4}$

a



b

$$\arg(z + 3 + 2i) = \frac{3\pi}{4}$$

$$\arg(x + iy + 3 + 2i) = \frac{3\pi}{4}$$

$$\arg((x + 3) + i(y + 2)) = \frac{3\pi}{4}$$

$$\frac{y + 2}{x + 3} = \tan\frac{3\pi}{4}$$

$$y + 2 = -(x + 3)$$

Hence the Cartesian equation of the locus is $y = -x - 5$, $x < -3$

$z + 3 + 2i$ can be written as $z - (-3 - 2i)$. As $\arg(z + 3 + 2i) = \frac{3\pi}{4}$, the locus of z is the half-line from $(-3, -2)$ making an angle of $\frac{3\pi}{4}$ in an anticlockwise sense from a line in the same direction as the positive real axis.

z can be rewritten as $z = x + iy$.

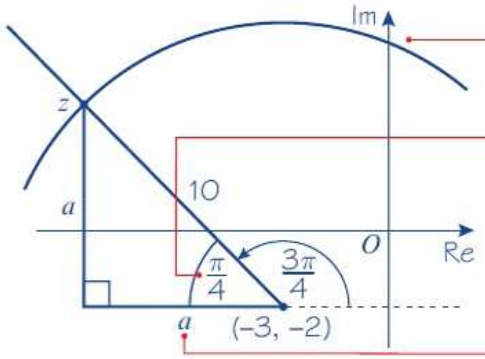
Group the real and imaginary parts.

Remove the argument.

$\tan\frac{3\pi}{4} = -1$

Watch out The locus is the half-line so you need to give a suitable range of values for x.

c $|z + 3 + 2i| = 10$ is a circle with centre $(-3, -2)$ and radius 10.



$$a^2 + a^2 = 10^2 \Rightarrow 2a^2 = 100$$

$$a = \sqrt{50} = \pm 5\sqrt{2}$$

$$z = (-3 - 5\sqrt{2}) + i(-2 + 5\sqrt{2})$$

Use a geometric approach to find z.

Draw part of a circle with centre $(-3, -2)$ and radius 10.

Angle inside the new triangle is $\pi - \frac{3\pi}{4} = \frac{\pi}{4}$

As the angle is $\frac{\pi}{4}$, the triangle is isosceles. So the two shorter sides have the same length.

Problem-solving An alternative algebraic approach would be to substitute the equation for the half-line, $y = -x - 5$, into the equation of the circle, $(x + 3)^2 + (y + 2)^2 = 10^2$, and then solve for x and y. You would need to choose the solution which lies on the correct half line.

Exercise 2E

(P) **1** Sketch the locus of z and give the Cartesian equation of the locus of z when:

a $|z| = 6$ **b** $|z| = 10$ **c** $|z - 3| = 2$

d $|z + 3i| = 3$ **e** $|z - 4i| = 5$ **f** $|z + 1| = 1$

g $|z - 1 - i| = 5$ **h** $|z + 3 + 4i| = 4$ **i** $|z - 5 + 6i| = 5$

Hint You may choose a geometric or an algebraic approach to answer these questions.

2 Given that z satisfies $|z - 5 - 4i| = 8$,

a sketch the locus of z on an Argand diagram

b find the exact values of z that satisfy:

i both $|z - 5 - 4i| = 8$ and $\text{Re}(z) = 0$ **ii** both $|z - 5 - 4i| = 8$ and $\text{Im}(z) = 0$

(P) **3** A complex number z is represented by the point P on the Argand diagram.

Given that $|z - 5 + 7i| = 5$,

a sketch the locus of P

b find the Cartesian equation of this locus

c find the maximum value of $\arg z$ in the interval $(-\pi, \pi)$.

(E/P) **4** On an Argand diagram the point P represents the complex number z.

Given that $|z - 4 - 3i| = 8$,

a find the Cartesian equation for the locus of P **(2 marks)**

b sketch the locus of P **(2 marks)**

c find the maximum and minimum values of $|z|$ for points on this locus. **(2 marks)**

(E/P) **5** The point P represents a complex number z on an Argand diagram.

Given that $|z + 2 - 2\sqrt{3}i| = 2$,

a sketch the locus of P on an Argand diagram **(2 marks)**

b write down the minimum value of $\arg z$ **(2 marks)**

c find the maximum value of $\arg z$. **(2 marks)**

6 Sketch the locus of z and give the Cartesian equation of the locus of z when:

a $|z - 6| = |z - 2|$ **b** $|z + 8| = |z - 4|$

c $|z| = |z + 6i|$ **d** $|z + 3i| + |z - 8i|$

e $|z - 2 - 2i| = |z + 2 + 2i|$ **f** $|z + 4 + i| = |z + 4 + 6i|$

g $|z + 3 - 5i| = |z - 7 - 5i|$ **h** $|z + 4 - 2i| = |z - 8 + 2i|$

i $\dfrac{|z + 3|}{|z - 6i|} = 1$ **j** $\dfrac{|z + 6 - i|}{|z - 10 - 5i|} = 1$

E/P **7** Given that $|z - 3| = |z - 6i|$,

a sketch the locus of z **(3 marks)**

b find the exact least possible value of $|z|$. **(4 marks)**

E/P **8** Given that $|z + 3 + 3i| = |z - 9 - 5i|$,

a sketch the locus of z **(3 marks)**

b find the Cartesian equation of this locus **(3 marks)**

c find the exact least possible value of $|z|$. **(3 marks)**

9 Sketch the locus of z and give the Cartesian equation of the locus of z when:

a $|2 - z| = 3$ **b** $|5i - z| = 4$ **c** $|3 - 2i - z| = 3$

10 Sketch the locus of z when:

a $\arg z = \frac{\pi}{3}$ **b** $\arg(z + 3) = \frac{\pi}{4}$ **c** $\arg(z - 2) = \frac{\pi}{2}$

d $\arg(z + 2 + 2i) = -\frac{\pi}{4}$ **e** $\arg(z - 1 - i) = \frac{3\pi}{4}$ **f** $\arg(z + 3i) = \pi$

g $\arg(z - 1 + 3i) = \frac{2\pi}{3}$ **h** $\arg(z - 3 + 4i) = -\frac{\pi}{2}$ **i** $\arg(z - 4i) = -\frac{3\pi}{4}$

P **11** Given that z satisfies $|z + 2i| = 3$,

a sketch the locus of z on an Argand diagram

b find $|z|$ that satisfies both $|z + 2i| = 3$ and $\arg z = \frac{\pi}{6}$

E/P **12** Given that the complex number z satisfies the equation $|z + 6 + 6i| = 4$,

a find the exact maximum and minimum value of $|z|$ **(3 marks)**

b find the range of values for θ, $-\pi < \theta < \pi$, for which $\arg(z - 4 + 2i) = \theta$ and $|z + 6 + 6i| = 4$ have no common solutions. **(4 marks)**

E/P **13** The point P represents a complex number z on an Argand diagram such that $|z| = 5$.
The point Q represents a complex number z on an Argand diagram such that $\arg(z + 4) = \frac{\pi}{2}$

a Sketch, on the same Argand diagram, the locus of P and the locus of Q as z varies. **(2 marks)**

b Find the complex number for which both $|z| = 5$ and $\arg(z + 4) = \frac{\pi}{2}$ **(2 marks)**

E/P **14** Given that the complex number z satisfies $|z - 2 - 2i| = 2$,

a sketch, on an Argand diagram, the locus of z **(2 marks)**

Given further that $\arg(z - 2 - 2i) = \frac{\pi}{6}$,

b find the value of z in the form $a + ib$, where $a, b \in \mathbb{R}$. **(4 marks)**

E/P **15** Sketch on the same Argand diagram the locus of points satisfying:

a $|z - 2i| = |z - 8i|$ **(2 marks)**

b $\arg(z - 2 - i) = \frac{\pi}{4}$ **(3 marks)**

The complex number z satisfies both $|z - 2i| = |z - 8i|$ and $\arg(z - 2 - i) = \frac{\pi}{4}$ **(2 marks)**

c Use your answers to parts **a** and **b** to find the value of z.

E/P **16** Sketch on the same Argand diagram the locus of points satisfying:

a $|z - 3 + 2i| = 4$ **(2 marks)**

b $\arg(z - 1) = -\frac{\pi}{4}$ **(3 marks)**

The complex number z satisfies both $|z - 3 + 2i| = 4$ and $\arg(z - 1) = -\frac{\pi}{4}$

Given that $z = a + ib$, where $a, b \in \mathbb{R}$,

c find the exact value of a and the exact value of b. **(3 marks)**

E/P **17** If the complex number z satisfies both $\arg z = \frac{\pi}{3}$ and $\arg(z - 4) = \frac{\pi}{2}$,

a find the value of z in the form $a + ib$, where $a, b \in \mathbb{R}$. **(3 marks)**

b Hence, find $\arg(z - 8)$. **(2 marks)**

E/P **18** Given that $\arg(z + 4) = \frac{\pi}{3}$,

a sketch the locus of z on an Argand diagram **(3 marks)**

b find the minimum value of $|z|$ for points on this locus. **(2 marks)**

E/P **19** A complex number z is represented by the point P on the Argand diagram. Given $|z + 8 - 4i| = 2$,

a sketch the locus of P **(2 marks)**

b show that the maximum value of $\arg(z + 15 - 2i)$ in the interval $(-\pi, \pi)$ is $2\arcsin\left(\frac{2}{\sqrt{53}}\right)$ **(3 marks)**

c find the exact values of the complex numbers that satisfy both $|z + 8 - 4i| = 2$ and $\arg(z + 4i) = \frac{3\pi}{4}$ **(3 marks)**

Challenge

The complex number z satisfies both $|z + i| = 5$ and $\arg(z - 2i) = \theta$, where θ is a real constant such that $-\pi < \theta \leqslant \pi$.
Given that $|z - 4i| < 3$, find the range of possible values of θ.

2.5 Regions in the Argand diagram

You can use complex numbers to represent regions on an Argand diagram.

Example 16

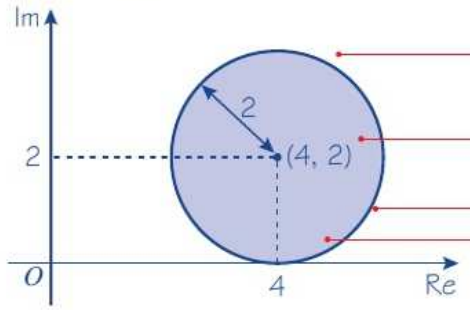
a On separate Argand diagrams, shade in the regions represented by:

i $|z - 4 - 2i| \leqslant 2$ **ii** $|z - 4| < |z - 6|$ **iii** $0 \leqslant \arg(z - 2 - 2i) \leqslant \frac{\pi}{4}$

b Hence, on the same Argand diagram, shade the region which satisfies

$\{z \in \mathbb{C} : |z - 4 - 2i| \leqslant 2\} \cap \{z \in \mathbb{C} : |z - 4| < |z - 6|\} \cap \left\{z \in \mathbb{C} : 0 \leqslant \arg(z - 2 - 2i) \leqslant \frac{\pi}{4}\right\}$

a i $|z - 4 - 2i| \leqslant 2$

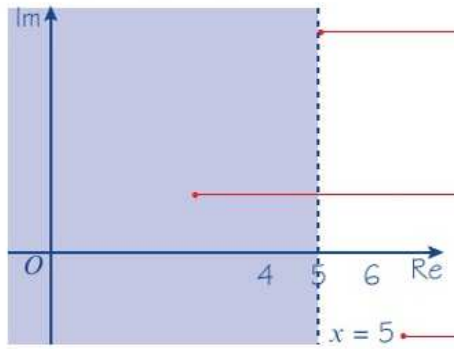


$|z - 4 - 2i| = 2$ represents a circle centre (4, 2), radius 2.

$|z - 4 - 2i| < 2$ represents the region on the inside of this circle.

$|z - 4 - 2i| \leqslant 2$ represents the boundary inside of this circle.

ii $|z - 4| < |z - 6|$

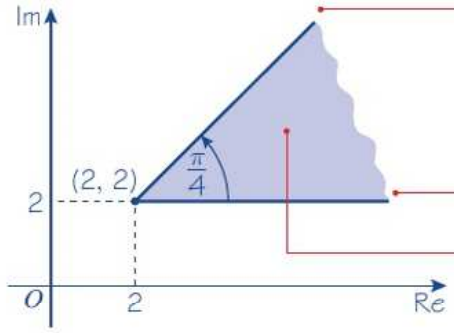


$|z - 4| = |z - 6|$ is represented by the line $x = 5$. This line is the perpendicular bisector of the line segment joining (4, 0) to (6, 0).

$|z - 4| < |z - 6|$ represents the region $x < 5$. All points in this region are closer to (4, 0) than to (6, 0).

Note this region does not include the line $x = 5$. So $x = 5$ is represented by a dashed line.

iii $0 \leqslant \arg(z - 2 - 2i) \leqslant \frac{\pi}{4}$



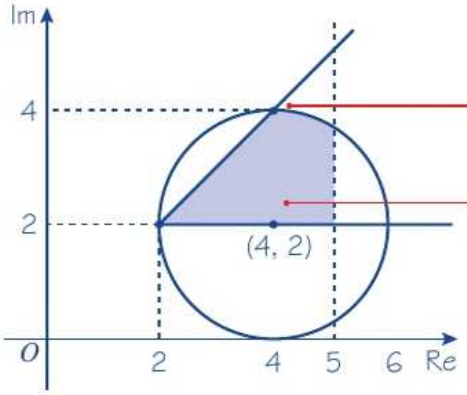
$\arg(z - 2 - 2i) = \frac{\pi}{4}$ is the half-line from the point (2, 2) at angle $\frac{\pi}{4}$ to the horizontal.

$\arg(z - 2 - 2i) = 0$ is the other half-line shown from the point (2, 2).

$0 \leqslant \arg(z - 2 - 2i) \leqslant \frac{\pi}{4}$ is represented by the region in between and including these two half-lines.

Notation The symbol ∩ is the symbol for the **intersection** of two sets. You need to find the region of points that lie in all three sets.

b $|z - 4 - 2i| \leqslant 2$, $|z - 4| < |z - 6|$ and $0 \leqslant \arg(z - 2 - 2i) \leqslant \frac{\pi}{4}$



The line $\arg(z - 2 - 2i) = \frac{\pi}{4}$ and the circle $|z - 4 - 2i| = 2$ both go through the point (4, 4).

The region shaded is satisfied by all three of
$|z - 4 - 2i| \leqslant 2$
$|z - 4| < |z - 6|$
$0 \leqslant \arg(z - 2 - 2i) \leqslant \frac{\pi}{4}$

Online Explore this region using GeoGebra.



Exercise 2F

1 On an Argand diagram, shade in the regions represented by the following inequalities:

a $|z| < 3$ **b** $|z - 2i| > 2$ **c** $|z + 7| \geqslant |z - 1|$ **d** $|z + 6| > |z + 2 + 8i|$

e $2 \leqslant |z| \leqslant 3$ **f** $1 \leqslant |z + 4i| \leqslant 4$ **g** $3 \leqslant |z - 3 + 5i| \leqslant 5$

E/P **2** The region R in an Argand diagram is satisfied by the inequalities $|z| \leqslant 5$ and $|z| \leqslant |z - 6i|$.
Draw an Argand diagram and shade in the region R. **(6 marks)**

E/P **3** The complex number z is represented by a point P on an Argand diagram.
Given that $|z + 1 - i| \leqslant 1$ and $0 \leqslant \arg z \leqslant \frac{3\pi}{4}$, shade the locus of P. **(6 marks)**

E/P **4** Shade on an Argand diagram the region satisfied by

$$\{z \in \mathbb{C} : |z| \leqslant 3\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{4} \leqslant \arg(z + 3) \leqslant \pi\right\}$$ **(6 marks)**

E/P **5 a** Sketch on the same Argand diagram:

i the locus of points representing $|z - 2| = |z - 6 - 8i|$ **(2 marks)**

ii the locus of points representing $\arg(z - 4 - 2i) = 0$ **(2 marks)**

iii the locus of points representing $\arg(z - 4 - 2i) = \frac{\pi}{2}$ **(2 marks)**

b Shade on an Argand diagram the set of points

$$\{z \in \mathbb{C} : |z - 2| \leqslant |z - 6 - 8i|\} \cap \left\{z \in \mathbb{C} : 0 \leqslant \arg(z - 4 - 2i) \leqslant \frac{\pi}{2}\right\}$$ **(2 marks)**

E/P **6 a** Find the Cartesian equations of:

i the locus of points representing $|z + 10| = |z - 6 - 4i\sqrt{2}|$

ii the locus of points representing $|z + 1| = 3$. **(6 marks)**

b Find the two values of z that satisfy both $|z + 10| = |z - 6 - 4i\sqrt{2}|$ and $|z + 1| = 3$. **(2 marks)**

c Hence shade in the region R on an Argand diagram which satisfies both $|z + 10| \leqslant |z - 6 - 4i\sqrt{2}|$ and $|z + 1| \leqslant 3$. **(4 marks)**

Challenge

The sets A, B and C are defined as:

$A = \{z \in \mathbb{C} : |z + 5 + 8i| \leqslant 5\}$

$B = \{z \in \mathbb{C} : |z + 8 + 4i| \leqslant |z + 2 + 12i|\}$

$C = \left\{z \in \mathbb{C} : 0 \leqslant \arg(z + 10 + 8i) \leqslant \frac{\pi}{4}\right\}$

Shade the set of points $A \cap B \cap C'$, that are in set A and in set B, but not in set C.

Mixed exercise 2

(E) **1** $f(z) = z^2 + 5z + 10$

a Find the roots of the equation $f(z) = 0$, giving your answers in the form $a \pm ib$, where a and b are real numbers. **(3 marks)**

b Show these roots on an Argand diagram. **(1 mark)**

(E/P) **2** $f(z) = z^3 + z^2 + 3z - 5$

Given that $f(-1 + 2i) = 0$,

a find all the solutions to the equation $f(z) = 0$ **(4 marks)**

b show all the roots of $f(z) = 0$ on a single Argand diagram **(2 marks)**

c prove that these three points are the vertices of a right-angled triangle. **(2 marks)**

(E/P) **3** $f(z) = z^4 - z^3 + 13z^2 - 47z + 34$

Given that $z = -1 + 4i$ is a solution to the equation,

a find all the solutions to the equation $f(z) = 0$ **(4 marks)**

b show all the roots on a single Argand diagram. **(2 marks)**

(E) **4** The real and imaginary parts of the complex number $z = x + iy$ satisfy the equation

$$(4 - 3i)x - (1 + 6i)y - 3 = 0$$

a Find the value of x and the value of y. **(3 marks)**

b Show z on an Argand diagram. **(1 mark)**

Find the values of:

c $|z|$ **(2 marks)**

d $\arg z$ **(2 marks)**

(E) **5** $z_1 = 4 + 2i$, $z_2 = -3 + i$

a Draw points representing z_1 and z_2 on the same Argand diagram. **(1 mark)**

b Find the exact value of $|z_1 - z_2|$. **(2 marks)**

Given that $w = \dfrac{z_1}{z_2}$,

c express w in the form $a + ib$, where $a, b \in \mathbb{R}$ **(2 marks)**

d find $\arg w$, giving your answer in radians. **(2 marks)**

(E) **6** A complex number z is given by $z = a + 4i$ where a is a non-zero real number.

a Find $z^2 + 2z$ in the form $x + iy$, where x and y are real expressions in terms of a. **(4 marks)**

Given that $z^2 + 2z$ is real,

b find the value of a. **(1 mark)**

Using this value for a,

c find the values of the modulus and argument of z, giving the argument in radians and giving your answers correct to 3 significant figures. **(3 marks)**

d Show the complex numbers z, z^2 and $z^2 + 2z$ on a single Argand diagram. **(3 marks)**

E **7** The complex number z is defined by $z = \dfrac{3 + 5\mathrm{i}}{2 - \mathrm{i}}$

Find:

a $|z|$ **(4 marks)**

b $\arg z$ **(2 marks)**

E **8** $z = 1 + 2\mathrm{i}$

a Show that $|z^2 - z| = 2\sqrt{5}$. **(4 marks)**

b Find $\arg(z^2 - z)$, giving your answer in radians to 2 decimal places. **(2 marks)**

c Show z and $z^2 - z$ on a single Argand diagram. **(2 marks)**

E **9** $z = \dfrac{1}{2 + \mathrm{i}}$

a Express in the form $a + b\mathrm{i}$, where $a, b \in \mathbb{R}$,

i z^2 **ii** $z - \dfrac{1}{z}$ **(4 marks)**

b Find $|z^2|$. **(2 marks)**

c Find $\arg\left(z - \dfrac{1}{z}\right)$, giving your answer in radians to two decimal places. **(2 marks)**

E/P **10** $z = \dfrac{a + 3\mathrm{i}}{2 + a\mathrm{i}}, \quad a \in \mathbb{R}$

a Given that $a = 4$, find $|z|$.

b Show that there is only one value of a for which $\arg z = \dfrac{\pi}{4}$, and find this value.

E **11** $z_1 = -1 - \mathrm{i}$, $z_2 = 1 + \mathrm{i}\sqrt{3}$

a Express z_1 and z_2 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $-\pi < \theta \leqslant \pi$. **(2 marks)**

b Find the modulus of:

i $z_1 z_2$ **ii** $\dfrac{z_1}{z_2}$ **(2 marks)**

c Find the argument of:

i $z_1 z_2$ **ii** $\dfrac{z_1}{z_2}$ **(2 marks)**

E **12** $z = 2 - 2\mathrm{i}\sqrt{3}$

Find:

a $|z|$ **(1 mark)**

b $\arg z$, in terms of π. **(2 marks)**

$w = 4\left(\cos\left(-\frac{\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{\pi}{4}\right)\right)$

Find:

c $\left|\dfrac{w}{z}\right|$ **(1 mark)**

d $\arg\left(\dfrac{w}{z}\right)$, in terms of π. **(2 marks)**

E **13** Express $4 - 4\mathrm{i}$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $r > 0$, $-\pi < \theta \leqslant \pi$, giving r and θ as exact values. **(3 marks)**

E/P **14** The point P represents a complex number z in an Argand diagram.
Given that $|z + 1 - i| = 1$,

a find a Cartesian equation for the locus of P **(2 marks)**

b sketch the locus of P on an Argand diagram **(2 marks)**

c find the greatest and least possible values of $|z|$ **(2 marks)**

d find the greatest and least possible values of $|z - 1|$. **(2 marks)**

P **15** Given that $\arg(z - 2 + 4i) = \frac{\pi}{4}$,

a sketch the locus of $P(x, y)$ which represents z on an Argand diagram

b find the minimum value of $|z|$ for points on this locus.

E/P **16** The complex number z satisfies $|z + 3 - 6i| = 3$. Show that the exact maximum value of $\arg z$ in the interval $(-\pi, \pi)$ is $\frac{\pi}{2} + 2\arcsin\left(\frac{1}{\sqrt{5}}\right)$. **(4 marks)**

E/P **17** A complex number z is represented by the point P on the Argand diagram.
Given that $|z - 5| = 4$,

a sketch the locus of P. **(2 marks)**

b Find the complex numbers that satisfy both $|z - 5| = 4$ and $\arg(z + 3i) = \frac{\pi}{3}$, giving your answers in radians to 2 decimal places. **(6 marks)**

c Given that $\arg(z + 5) = \theta$ and $|z - 5| = 4$ have no common solutions, find the range of possible values of θ, $-\pi < \theta < \pi$. **(3 marks)**

E/P **18** Given that $|z + 5 - 5i| = |z - 6 - 3i|$,

a sketch the locus of z **(3 marks)**

b find the Cartesian equation of this locus **(3 marks)**

c find the least possible value of $|z|$. **(3 marks)**

E/P **19** **a** Find the Cartesian equation of the locus of points that satisfies $|z - 4| = |z - 8i|$. **(3 marks)**

b Find the value of z that satisfies both $|z - 2| = |z - 4i|$ and $\arg z = \frac{\pi}{4}$ **(3 marks)**

c Shade on an Argand diagram the set of points
$\{z \in \mathbb{C} : |z - 4| \leqslant |z - 8i|\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{4} \leqslant \arg z \leqslant \pi\right\}$ **(3 marks)**

E/P **20** **a** Find the Cartesian equations of:

i the locus of points representing $|z - 3 + i| = |z - 1 - i|$

ii the locus of points representing $|z - 2| = 2\sqrt{2}$. **(6 marks)**

b Find the two values of z that satisfy both $|z - 3 + i| = |z - 1 - i|$ and $|z - 2| = 2\sqrt{2}$. **(2 marks)**

The region R is defined by the inequalities $|z - 3 + i| \geqslant |z - 1 - i|$ and $|z - 2| \leqslant 2\sqrt{2}$.

c Show the region R on an Argand diagram. **(4 marks)**

Challenge

The complex number z satisfies $\arg(z - 3 + 3i) = -\frac{\pi}{4}$
The complex number w is such that $|w - z| = 3$.

a Sketch the locus of w.

b State the exact minimum value of $|w|$.

Summary of key points

1 You can represent complex numbers on an **Argand diagram**. The x-axis on an Argand diagram is called the **real axis** and the y-axis is called the **imaginary axis**. The complex number $z = x + iy$ is represented on the diagram by the point $P(x, y)$, where x and y are Cartesian coordinates.

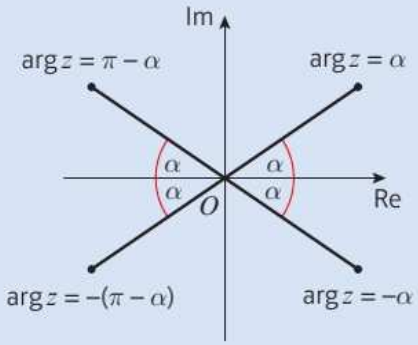
2 The complex number $z = x + iy$ can be represented as the vector $\begin{pmatrix} x \\ y \end{pmatrix}$ on an Argand diagram.

3 The **modulus** of a complex number, $|z|$, is the distance from the origin to that number on an Argand diagram. For a complex number $z = x + iy$, the modulus is given by $|z| = \sqrt{x^2 + y^2}$.

4 The **argument** of a complex number, $\arg z$, is the angle between the positive real axis and the line joining that number to the origin on an Argand diagram. For a complex number $z = x + iy$, the argument, θ, satisfies $\tan\theta = \frac{y}{x}$

5 Let α be the positive acute angle made with the real axis by the line joining the origin and z.

- If z lies in the first quadrant then $\arg z = \alpha$.
- If z lies in the second quadrant then $\arg z = \pi - \alpha$.
- If z lies in the third quadrant then $\arg z = -(\pi - \alpha)$.
- If z lies in the fourth quadrant then $\arg z = -\alpha$.



6 For a complex number z with $|z| = r$ and $\arg z = \theta$, the modulus–argument form of z is $z = r(\cos\theta + i\sin\theta)$

7 For any two complex numbers z_1 and z_2,

- $|z_1 z_2| = |z_1||z_2|$
- $\arg(z_1 z_2) = \arg z_1 + \arg z_2$
- $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$
- $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$

8 For two complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, $|z_2 - z_1|$ represents the distance between the points z_1 and z_2 on an Argand diagram.

9 Given $z_1 = x_1 + iy_1$, the locus of points z on an Argand diagram such that $|z - z_1| = r$, or $|z - (x_1 + iy_1)| = r$, is a circle with centre (x_1, y_1) and radius r.

10 Given $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, the locus of points z on an Argand diagram such that $|z - z_1| = |z - z_2|$ is the perpendicular bisector of the line segment joining z_1 and z_2.

11 Given $z_1 = x_1 + iy_1$, the locus of points z on an Argand diagram such that $\arg(z - z_1) = \theta$ is a half-line from, but not including, the fixed point z_1 making an angle θ with a line from the fixed point z_1 parallel to the real axis.